

Continual Learning for fMRI-Based Brain Disorder Diagnosis via Functional Connectivity Matrices Generative Replay

Qianyu Chen[†]

[†]Nanyang Technological University
 CHEN1981@e.ntu.edu.sg

Shujian Yu^{‡,§}

[‡]VU Amsterdam; [§]UiT The Arctic University of Norway
 yusj9011@gmail.com

Abstract

Functional magnetic resonance imaging (fMRI) is widely used for studying and diagnosing brain disorders, with functional connectivity (FC) matrices providing powerful representations of large-scale neural interactions. However, existing diagnostic models are trained either on a single site or under full multi-site access, making them unsuitable for real-world scenarios where clinical data arrive sequentially from different institutions. This results in limited generalization and severe catastrophic forgetting. This paper presents the first continual learning framework specifically designed for fMRI-based diagnosis across heterogeneous clinical sites. Our framework introduces a structure-aware variational autoencoder that synthesizes realistic FC matrices for both patient and control groups. Built on this generative backbone, we develop a multi-level knowledge distillation strategy that aligns predictions and graph representations between new-site data and replayed samples. To further enhance efficiency, we incorporate a hierarchical contextual bandit scheme for adaptive replay sampling. Experiments on multi-site datasets for major depressive disorder (MDD), schizophrenia (SZ), and autism spectrum disorder (ASD) show that the proposed generative model enhances data augmentation quality, and the overall continual learning framework substantially outperforms existing methods in mitigating catastrophic forgetting. Our code is available at <https://github.com/4me808/FORGE>.

1. Introduction

Functional magnetic resonance imaging (fMRI) is a non-invasive technique that measures brain activity through blood-oxygen-level-dependent (BOLD) signals. Functional connectivity (FC) matrices derived from these signals encode pairwise correlations between brain regions, characterizing large-scale network organization and inter-regional interaction patterns. As such, FC provides a principled repre-

sentation of brain network topology and a powerful modality for probing neurobiological mechanisms and disease-related alterations [8, 11, 31]. FC matrices, combined with graph-based modeling techniques [8, 16], have been widely employed for brain disorder diagnosis. However, existing fMRI-based diagnostic models still exhibit limited reliability for practical clinical deployment [1, 7, 31]. Clinical data typically arrive sequentially from different sites, yet most models are trained offline on a single dataset or assume immediate access to all data. Consequently, they lack mechanisms for continual generalization to newly arriving data and suffer from the overwriting of previously learned information, a phenomenon known as *catastrophic forgetting* [30, 41]. When such models are adapted to new sites, updates to accommodate new distributions often overwrite prior knowledge, substantially undermining their clinical utility in real-world deployment [7, 20, 29]. This limitation motivates the development of continual learning (CL) frameworks that can integrate new-site data incrementally while preserving previously acquired knowledge [1, 30].

Various strategies have been proposed to preserve knowledge from previous tasks in CL, with notable examples including Elastic Weight Consolidation (EWC) [20] and PackNet [28]. Among them, rehearsal-based approaches alleviate forgetting by explicitly storing and replaying a few representative samples from previous tasks in a memory buffer. In healthcare, however, experience replay is often constrained by privacy regulations that prohibit sharing raw patient data across institutions [5, 7, 12]. Furthermore, most CL research has focused on image-based tasks, leaving graph-structured medical data, especially fMRI, largely underexplored [8, 23, 30, 41, 54]. These considerations highlight the urgent need for a graph-specific, privacy-friendly CL framework tailored for streaming fMRI data.

In this paper, we propose FORGE (Functional cOnnectivity Replay with Graph rEpresentation), a CL framework tailored for FC matrices collected across multiple clinical sites. FORGE addresses catastrophic forgetting by replaying both low-level FC matrices and higher-level knowledge in the form of classification logits and graph embeddings

within a unified knowledge distillation framework. A central contribution of this work is FCM-VAE, a novel generative model specifically designed for FC matrices. FCM-VAE enables privacy-preserving generative replay by synthesizing realistic FC patterns without exposing raw patient data. Unlike existing VAEs, it employs a structure-aware graph transformer encoder with local adjacency encoding and spectral positional encoding to capture the intrinsic topological and spectral properties of FC networks. Moreover, we introduce a low-rank decoder, which is well suited for FC matrices because large-scale functional connectivity is known to exhibit strong low-rank structure [35].

Our main contributions include:

- We propose FORGE, the first CL framework tailored for cross-site fMRI-based disease diagnosis under realistic streaming and privacy-friendly settings. FORGE unifies dual-level knowledge distillation with generative replay.
- We develop FCM-VAE, a conditional variational autoencoder specifically designed for FC matrices. FCM-VAE synthesizes realistic and diverse brain network patterns by integrating local adjacency encoding, spectral positional encoding, and a low-rank decoder, and it outperforms existing graph generative models in sample fidelity.
- Extensive experiments on large-scale neuroimaging datasets, including ABIDE [8], REST-meta-MDD [44], and BSNIP [6], show that FCM-VAE significantly improves augmentation quality, and FORGE consistently exceeds state-of-the-art baselines in predictive accuracy and resistance to catastrophic forgetting.

2. RELATED WORK

2.1. Continual Learning and its Extensions to Graph Data

Existing continual learning (CL) approaches generally fall into four main categories. Regularization-based methods constrain parameter updates or enforce consistency with earlier predictions to preserve prior knowledge [20, 24]. Parameter-isolation methods allocate task-specific model capacity to reduce interference and forgetting [28, 34, 46]. Rehearsal-based strategies replay real or synthetic samples from previous tasks to maintain performance over time [2, 4, 17, 32]. Knowledge-distillation methods adopt a teacher–student paradigm, guiding the current model to align with the outputs or representations of earlier models [37, 47, 50]. However, most CL frameworks are developed for image-based data. When applied to graph-structured domains, they often struggle with non-Euclidean topologies and evolving relational structures, limiting their ability to prevent forgetting and maintain generalization [39].

Recent work has begun to develop CL frameworks for graph data [51, 53]. These approaches typically extend

conventional CL paradigms with graph-aware mechanisms such as structure-guided replay, topology-regularized updates, and subgraph sampling, aiming to improve node representations and retain knowledge over time [21, 49, 53, 55]. However, most existing studies focus on node-level classification, while graph-level classification remains largely underexplored [39]. A few works address graph continual classification using basic rehearsal strategies that replay stored graphs to alleviate forgetting [3, 14], or by preserving topology-aware features to reduce representation drift [25, 52]. Notably, none of these methods have been extended to or evaluated on brain network classification tasks.

2.2. Continual Learning for Brain Signals

Several recent studies have begun exploring CL for neural recording signals, but the literature has largely focused on subject-specific adaptation for EEG. For example, Shi et al. [36] proposes a memory projection strategy that aligns gradients between new and old tasks through dynamic projection matrices, reducing forgetting in seizure prediction. Duan et al. [9] introduces a balanced replay buffer that maintains a small class-balanced subset of samples to alleviate forgetting in online EEG decoding. More recently, Zhou et al. [56] presents BrainUICL, an unsupervised framework that adapts to new subjects without labels while preserving performance on previously observed subjects. In contrast, our work addresses a substantially more challenging cross-site setting for fMRI, where heterogeneity across institutions, scanners, and acquisition protocols introduces far greater distributional shift. Our goal is to mitigate forgetting under this site-level variability while ensuring accuracy, generalization, and privacy preservation in practical deployment. From a neuroscience perspective, fMRI and EEG differ markedly in their spatial resolution, temporal dynamics [27], and network-level representations, implying fundamentally different challenges for CL.

3. FORGE

3.1. Problem Setup

We first describe how resting-state fMRI (rs-fMRI) data are transformed into FC matrices for classification, and then formalize the diagnosis task under CL.

rs-fMRI Data Processing Pipeline: For each subject, we preprocess the raw rs-fMRI scans following standard neuroimaging protocols. The brain is divided into 116 pre-defined regions of interest (ROIs) using the AAL-116 atlas [40]. For every ROI, we extract the blood-oxygen-level-dependent (BOLD) time series and compute Pearson correlation coefficients between all ROI pairs, yielding a symmetric 116×116 functional connectivity (FC) matrix. For subject i , we construct an undirected weighted graph $G_i = (\mathcal{V}_i, \mathcal{E}_i)$, where $|\mathcal{V}_i| = 116$ nodes correspond to ROIs,

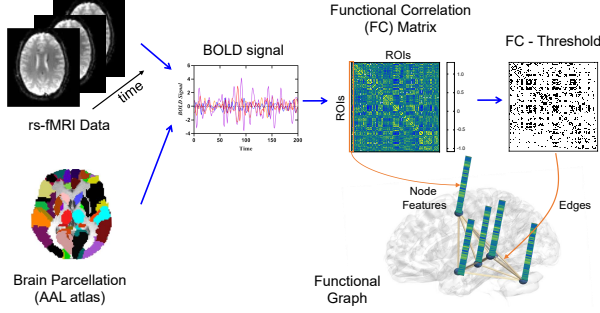


Figure 1. Pipeline for transforming raw rs-fMRI data into brain networks for the classification task.

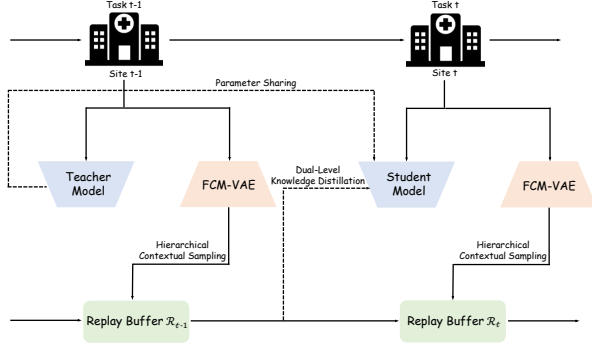


Figure 2. The continual learning workflow of FORGE across consecutive sites. At site $t-1$, FCM-VAE (Section 3.3) is trained to generate functional connectivity (FC) matrices, which are stored in a replay buffer. When moving to site t , a student classifier is trained to align with the previously learned model from site $t-1$, matching both graph-level representations and classification logits using data from the current site and replayed samples selected via the hierarchical contextual sampling strategy (Section 3.4). After training, a new FCM-VAE is updated at site t to produce site-specific synthetic samples for future learning.

and edges $\mathcal{E}_i$ are obtained by thresholding the FC matrix at $\tau = 0.4$. Each node feature is defined as the correlation profile of that ROI with all other ROIs, forming a 116-dimensional feature vector. A complete visualization of the preprocessing pipeline is provided in Fig. 1.

Continual Learning Setup: Let $\mathcal{M} = \{M_1, \dots, M_T\}$ denote T clinical sites. Each site M_t provides its own dataset, which we split into a training set $\mathcal{D}_t^{\text{train}} = \{(G_i^t, y_i^t)\}_{i=1}^{N_t^{\text{train}}}$ of size N_t^{train} and a test set $\mathcal{D}_t^{\text{test}} = \{(G_j^t, y_j^t)\}_{j=1}^{N_t^{\text{test}}}$ of size N_t^{test} , where $G_i^t \in \mathcal{G}$ is the functional connectivity graph for subject i and $y_i^t \in \mathcal{Y}$ is the diagnostic label. Tasks (sites) arrive sequentially. During training on task t , the model has access only to $\mathcal{D}_t^{\text{train}}$ and cannot revisit $\{\mathcal{D}_k^{\text{train}}\}_{k < t}$. After finishing the final site T , we evaluate performance on all test sets $\{\mathcal{D}_1^{\text{test}}, \dots, \mathcal{D}_T^{\text{test}}\}$ to assess cross-site generalization and forgetting. See Appendix A for details.

3.2. Dual-Level Knowledge Distillation Framework

Our framework integrates generative replay into a dual-level knowledge distillation scheme. A graph classifier f_θ is trained sequentially over tasks. For an input graph G , we denote the output logits by $h_\theta(G)$ and the corresponding predictive distribution by $f_\theta(G) = \text{softmax}(h_\theta(G))$.

At the current task t_c , we update the model parameters θ to perform well on $\mathcal{D}_{t_c}^{\text{train}}$ while matching the behavior of the frozen teacher model $f_{\theta_t^*}$ from earlier tasks $t < t_c$. Here, θ_t^* is the optimal set of parameters at the end of task t . Formally, we minimize:

$$\begin{aligned} & \mathbb{E}_{(G,y) \sim \mathcal{D}_{t_c}^{\text{train}}} [\ell(y, f_\theta(G))] + \lambda_1 \sum_{t=1}^{t_c-1} \mathbb{E}_{(G,y) \sim \mathcal{R}_t} [\ell(y, f_\theta(G))] \\ & + \lambda_2 \sum_{t=1}^{t_c-1} \mathbb{E}_{(G,u) \sim \mathcal{R}_t} [\|h_\theta(G) - u\|_2^2], \end{aligned} \quad (1)$$

where $\lambda_1, \lambda_2 > 0$, $u \triangleq h_{\theta_t^*}(G)$ denotes the stored teacher logits, and $\mathcal{R}_t$ is the replay buffer for task t . Note that replayed samples are not real FC matrices but synthetic data generated by FCM-VAE trained at each task t . The first term fits the current task, the second term uses generative replay to preserve performance on earlier tasks, and the third term aligns the student logits with the teacher logits.

Restricting alignment to the output logits overlooks discrepancies that may arise in intermediate representations. We further introduce a graph-readout distillation mechanism that directly aligns graph-level representations between the student and the teacher. For each graph G , we store the teacher readout $r \triangleq \rho(E_{\theta_t^*}(G))$, where $E_\theta(\cdot)$ is the graph encoder and $\rho(\cdot)$ is the readout function that aggregates node embeddings into a graph-level representation. The final unified objective becomes:

$$\begin{aligned} & \mathbb{E}_{(G,y) \sim \mathcal{D}_{t_c}^{\text{train}}} [\ell(y, f_\theta(G))] + \lambda_1 \sum_{t=1}^{t_c-1} \mathbb{E}_{(G,y) \sim \mathcal{R}_t} [\ell(y, f_\theta(G))] \\ & + \lambda_2 \sum_{t=1}^{t_c-1} \mathbb{E}_{(G,u) \sim \mathcal{R}_t} [\|h_\theta(G) - u\|_2^2] \\ & + \lambda_3 \sum_{t=1}^{t_c-1} \mathbb{E}_{(G,r) \sim \mathcal{R}_t} [\|\rho(E_\theta(G)) - r\|_2^2], \end{aligned} \quad (2)$$

where $\lambda_1, \lambda_2, \lambda_3 > 0$ are weighting coefficients.

For each site M_t , the replay buffer $\mathcal{R}_t$ stores tuples of the form (G, y, u, r) , where G is a synthesized graph with label y , $u = h_{\theta_t^*}(G)$ denotes the teacher logits, and $r = \rho(E_{\theta_t^*}(G))$ is the corresponding teacher graph-level representation. The teacher outputs u and r are computed once when the sample is added to the buffer and remain fixed thereafter, providing stable and consistent supervision during replay. See Fig. 2 for the entire framework.

3.3. Constructing Replay Buffer with FCM-VAE

We describe the construction of our VAE-based generative model (see Fig. 3) designed specifically for FC matrices.

3.3.1. Structure-aware Graph Transformer Encoder

To address the challenges of limited samples and high-dimensional FC matrices, we embed the topological structure and spectral geometry of each individual into the encoder for more robust and personalized representations. Concretely, we propose a Structure-aware Graph Transformer that integrates adjacency and spectral positional biases into the multi-head attention, thereby leveraging both local connectivity patterns and global geometric features. First, for each subject and their ROI node $p \in \{1, \dots, |\mathcal{V}|\}$, we augment the initial node feature vector x_p as:

$$x_p^{(0)} = [A_p^{\text{ROI}}, \phi_p, \text{deg}_p] \in \mathbb{R}^{|\mathcal{V}|+k+1}, \quad (3)$$

where $A_p^{\text{ROI}} \in \mathbb{R}^{|\mathcal{V}|}$ denotes the p -th column of the FC matrix, $\Phi \in \mathbb{R}^{|\mathcal{V}| \times k}$ is the spectral embedding matrix formed by the top k non-trivial Laplacian eigenvectors, and $\phi_p \in \mathbb{R}^k$ is the spectral coordinate of node p in Φ , and deg_p denotes the degree of node p , encoding its local reachability. We concatenate the $|\mathcal{V}|$ node features to form the initial feature matrix $X^{(0)} = [x_1^{(0)}; x_2^{(0)}; \dots; x_{|\mathcal{V}|}^{(0)}] \in \mathbb{R}^{|\mathcal{V}| \times (|\mathcal{V}|+k+1)}$. Next, we feed $X^{(0)}$ into a multi-head self-attention layer. For each head h , three learnable projection matrices W_h^Q , W_h^K , and $W_h^V \in \mathbb{R}^{(|\mathcal{V}|+k+1) \times d_h}$ are used to obtain Q_h , K_h , and $V_h \in \mathbb{R}^{|\mathcal{V}| \times d_h}$, respectively, where $d_h = (|\mathcal{V}| + k + 1)/H$ and H is the number of heads [45].

Each head further incorporates a local adjacency bias and a spectral positional bias into the attention scores. *Local Adjacency Encoding* leverages the adjacency matrix to enforce local topological constraints and guide attention towards neighboring nodes. The attention bias is defined as:

$$b_h^{\text{LAE}} = \tilde{A} = A^{\text{adj}} + I \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}, \quad (4)$$

where $A^{\text{adj}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ is the binary adjacency matrix and I adds self-loops so that each node includes itself when computing attention.

Spectral positional encoding focuses on capturing global geometric information and allows nodes with similar spectral coordinates to attend more closely to each other. The attention bias is defined as:

$$b_h^{\text{SGE}} = \frac{(\Phi W_h^\phi)(\Phi W_h^\phi)^\top}{\sqrt{d_h}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}, \quad (5)$$

where $W_h^\phi \in \mathbb{R}^{k \times d_h}$ is a learnable projection that maps the spectral embedding into the d_h dimensional subspace of head h .

By integrating both local adjacency encoding and spectral positional encoding as additive biases, we obtain the

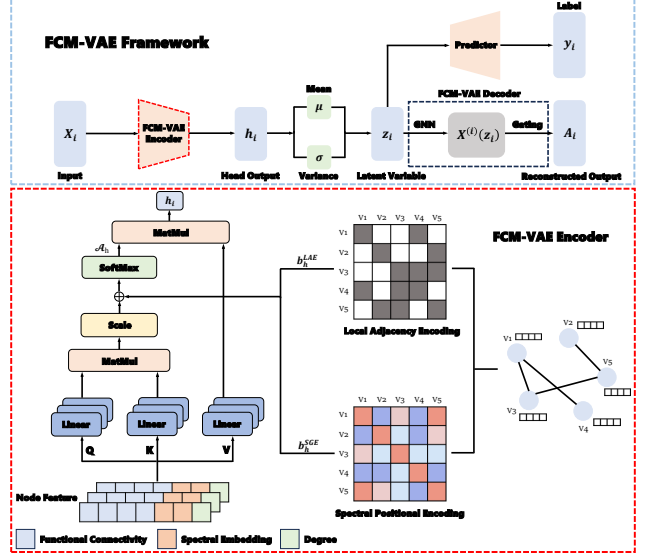


Figure 3. **Framework of FCM-VAE.** Given an input FC graph G , the encoder extracts a hidden representation h , which parameterizes a diagonal Gaussian distribution with mean μ and variance σ , from which the latent variable z is sampled. The decoder, implemented as a gated GNN, reconstructs the adjacency matrix A and predicts the phenotype y through a linear Gaussian supervision head applied directly to z . In the encoder, each node feature is formed by concatenating its functional connectivity profile, spectral embedding, and node degree. The attention layers incorporate both a local adjacency bias and a spectral positional bias, enabling the encoder to capture the intrinsic topological and spectral structure of FC networks.

final attention score:

$$\mathcal{H}_h = \frac{Q_h K_h^\top}{\sqrt{d_h}} + \tilde{A} + \frac{(\Phi W_h^\phi)(\Phi W_h^\phi)^\top}{\sqrt{d_h}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}. \quad (6)$$

Using additive biases preserves the expressiveness of full-graph attention. As a result, the proposed attention mechanism remains robust and expressive even in small-sample scenarios with complex graph structures.

From the attention score matrix, we obtain the attention weights $\mathcal{A}_h = \text{softmax}(\mathcal{H}_h)$, and head outputs $H_h = \mathcal{A}_h V_h$. Multi-head outputs are concatenated and passed through feed-forward and normalization layers for L layers to yield node representations $\{x_p^{(L)}\}_{p=1}^{|\mathcal{V}|}$. The graph-level embedding is obtained through a permutation-invariant readout function $h = \text{Readout}\left(\frac{1}{|\mathcal{V}|} \sum_{p=1}^{|\mathcal{V}|} x_p^{(L)}\right)$, where $\text{Readout}(\cdot)$ applies mean pooling over node representations followed by a linear projection.

We then employ h to parameterize the approximate posterior of the latent variable as a diagonal-covariance Gaussian distribution, following the design principle of VAE:

$$q_\phi(z | h) = \mathcal{N}(\mu_\phi(h), \text{diag}[\sigma_\phi^2(h)]), \quad z \in \mathbb{R}^{d_z} \quad (7)$$

where $\mu_\phi(h), \sigma_\phi^2(h) \in \mathbb{R}^{d_z}$ are encoder outputs that parameterize the mean and variance of the Gaussian posterior. This design preserves permutation invariance at the graph level while keeping the posterior parameterization simple.

3.3.2. Low-rank Decoder with Reachability Gating

On the decoder side, we reconstruct the functional connectivity matrix from the low-dimensional latent variable z . We assume that the reconstructed edge entries are conditionally independent given $z \in \mathbb{R}^{d_z}$ [26]. Accordingly, the likelihood of the reconstructed graph $\hat{A}$ factorizes as:

$$p_\psi(L(\hat{A}) = a \mid z) = \prod_{e=1}^{|\mathcal{V}|(|\mathcal{V}|-1)/2} p_\psi(\hat{A}_e = a_e \mid z), \quad (8)$$

where $p_\psi(\hat{A}_e \mid z)$ denotes the generative model for edge e , and we impose a Gaussian prior $p(z) = \mathcal{N}(0, I_{d_z})$.

To model connectivity strengths with positive support, we first transform each Pearson correlation entry $r_e \in [-1, 1]$ into a nonnegative intensity. Specifically, we apply the Fisher- z transform, standardize it using training-set statistics, and then exponentiate:

$$s_e = \text{atanh}(r_e), \quad \tilde{s}_e = \frac{s_e - \mu_s}{\sigma_s}, \quad x_e = \exp(\tilde{s}_e), \quad (9)$$

where μ_s and σ_s are the mean and standard deviation estimated from the training data. In Eq. (8), we therefore let a_e denote the transformed intensity x_e .

Given z , the decoder models each transformed edge intensity with a Poisson likelihood:

$$z \sim \mathcal{N}(0, I_{d_z}), \quad \hat{A}_e \mid z \sim \text{Poisson}(\hat{\lambda}_e(z)), \quad (10)$$

where the rate is parameterized with the canonical log link,

$$\hat{\lambda}_e(z) = \exp(\nu_e + \omega_e(z)), \quad (11)$$

where ν_e is an edge-specific baseline parameter shared across subjects, capturing the common connectivity tendency for edge e , while $\omega_e(z)$ models the subject-specific deviation induced by the latent representation z [26].

For edge $e = [u, v]$, the interaction term is defined as:

$$\omega_e(z) = \sum_{r=1}^R \alpha_r U_{ur}(z) U_{vr}(z), \quad (12)$$

where

$$U(z) = [U_{ur}(z)] \in \mathbb{R}^{|\mathcal{V}| \times R}, \quad (13)$$

$$U_r(z) = (U_{1r}(z), \dots, U_{|\mathcal{V}|r}(z))^\top = g_r(z).$$

Here, $U(z)$ is a decoder-generated node-factor matrix, α_r controls the contribution of the r -th latent factor, and $g_r(\cdot)$

is a nonlinear mapping parameterized by ψ . This factorization defines a low-rank bilinear decoder, in which each edge strength is reconstructed from the interaction of the two nodes across R latent factors [15]. Finally, all predicted edge intensities are assembled to form the reconstructed functional connectivity matrix $\hat{A}$.

Since FC matrices typically exhibit sparse individual-specific variations across subjects, we adopt a soft gating mechanism based on the adjacency matrix to constrain the strengths of the reconstructed edges, while avoiding spurious edge formation. We project the adjacency matrix of each subject into a gating matrix that modulates the decoder outputs:

$$G = \sigma(A^{\text{adj}} + \log(\varepsilon)), \quad A^{\text{pred}} = G \odot \hat{A}. \quad (14)$$

Here, $\sigma(\cdot)$ denotes the element-wise sigmoid function, $\odot$ is the Hadamard product, and ε is a small constant (e.g., 10^{-2}).

3.3.3. Linear Gaussian Supervised Head and Conditional Generation

To jointly reconstruct the graph and predict the phenotype y , we attach a linear-Gaussian supervision module on z . We posit a linear Gaussian relationship between the phenotype and the latent factors, leading to the following formulation:

$$p(y \mid z) = \mathcal{N}(\beta_0 + \beta^\top z, \sigma_y^2), \quad (15)$$

Here, β_0 and β are learnable parameters, and σ_y^2 denotes the observation noise variance. The linear Gaussian supervision head introduces minimal inductive bias by directly linking the label y to the latent variable, improving interpretability and calibration of the latent space. At inference, prediction is obtained from the posterior mean as $\hat{y} = \beta_0 + \beta^\top \mu_\phi(h)$, where $\mu_\phi(h)$ is the mean of the latent posterior given h .

3.3.4. Overall Training Objective

FCM-VAE is trained under a supervised evidence lower bound that jointly optimizes graph reconstruction and class prediction. The training objective is formulated as:

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\psi(A^{\text{ROI}} \mid z)] + \mathbb{E}_{q_\phi}[\log p_\theta(y \mid z)] - \beta D_{\text{KL}}(q_\phi(z \mid \cdot) \parallel p(z)) \quad (16)$$

where A^{ROI} and y denote the observed FC matrix and phenotype label of the current subject, and $\beta > 1$ serves as a regularization strength that encourages a compact latent representation and mitigates overfitting. The first term enforces accurate reconstruction between A^{pred} and A^{ROI} . The second term is a supervised likelihood that shapes the latent space so that z captures information predictive of the phenotype y . The third term encourages the variational posterior $q_\phi(z \mid \cdot)$ to remain close to the prior distribution $p(z)$, chosen as a standard Gaussian as in the conventional VAE.

3.4. Hierarchical Contextual Thompson Sampling

To make efficient use of a limited replay memory, we propose a Hierarchical Contextual Thompson Sampling (HCTS) framework that adaptively allocates replay capacity across hospital sites (site level) and selects the most informative samples within each site (sample level).

3.4.1. Site-Level Sampling

We assume a total replay capacity of K samples shared across all sites. The goal is to allocate k_i samples to each site M_i , such that $\sum_i k_i = K$. For each site, we define a context vector $\phi_i = [\text{Acc}_i, \text{Forget}_i]$, where Acc_i denotes current accuracy and Forget_i measures performance decay. The expected replay utility is modeled as:

$$r_i = \phi_i^\top \mathbf{w} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (17)$$

where $\mathbf{w}$ is the latent weight vector with posterior $p(\mathbf{w} \mid \mathcal{D})$. At each training round, Thompson Sampling draws a posterior sample $\tilde{\mathbf{w}}$ and estimates the utility:

$$\tilde{r}_i = \phi_i^\top \tilde{\mathbf{w}}. \quad (18)$$

The replay quota for each site is then determined by a normalized softmax allocation:

$$k_i = K \cdot \frac{\exp(\tilde{r}_i)}{\sum_j \exp(\tilde{r}_j)}, \quad (19)$$

ensuring $\sum_i k_i = K$. This mechanism prioritizes sites with higher uncertainty or forgetting, enhancing global stability.

3.4.2. Sample-Level Sampling

Given the site-specific quota k_i , we further identify the most beneficial samples from each buffer $\mathcal{R}_i$. Each sample $u \in \mathcal{R}_i$ is described by a context vector $\psi_u = [\text{margin}_u, \text{closeness}_u]$, capturing uncertainty and representativeness. The expected utility is modeled as:

$$r_u = \psi_u^\top \mathbf{v} + \epsilon_u, \quad \epsilon_u \sim \mathcal{N}(0, \sigma^2), \quad (20)$$

where $\mathbf{v}$ denotes the sample-level relevance weights. Thompson Sampling draws $\tilde{\mathbf{v}}$ from the posterior to compute:

$$\tilde{r}_u = \psi_u^\top \tilde{\mathbf{v}}. \quad (21)$$

The top- k_i samples with the highest $\tilde{r}_u$ values are selected with a greedy farthest-first traversal in the readout embedding space $\{\rho(E_\theta(G)) \mid G \in \mathcal{C}_i\}$, thereby improving coverage and reducing redundancy. The selection objective is formulated as:

$$\mathcal{S}_i = \arg \max_{S \subseteq \mathcal{C}_i: |S|=k_i} \min_{\substack{G, G' \in S \\ G \neq G'}} \|\rho(E_\theta(G)) - \rho(E_\theta(G'))\|_2. \quad (22)$$

This hierarchical design ensures replay focuses both on *where* forgetting occurs (site level) and *which* samples are most valuable for continual adaptation (sample level). We refer interested readers to Appendix C for details.

4. Experiments and Results

4.1. The Datasets and Preprocessing

To rigorously assess the proposed FORGE framework, we conduct experiments on three widely used, large-scale, multi-site fMRI datasets: the Autism Brain Imaging Data Exchange I (ABIDE-I) [8], the REST-meta-MDD dataset [44], and the Bipolar and Schizophrenia Network for Intermediate Phenotypes (BSNIP) [6]. The ABIDE-I dataset consists of rs-fMRI scans from 1099 participants across 17 independent sites. To ensure statistical reliability, we focus on the four largest sites in the ABIDE-I dataset: Site 6 (184 samples), Site 14 (99), Site 15 (145), and Site 16 (101). For the REST-meta-MDD dataset, the largest rs-fMRI resource for major depressive disorder (MDD), we follow the quality-control criteria [54] and select the five largest sites: Site 1 (146 samples), Site 6 (96), Site 10 (93), Site 14 (470), and Site 15 (144). The BSNIP contains 1,142 individuals, comprising 640 healthy controls and 502 participants diagnosed with schizophrenia. Similarly, we retain the five largest sites: Hartford (297 samples), Chicago (239), Dallas (197), Georgia (152), and Baltimore (120). All three datasets were preprocessed using a standard rs-fMRI pipeline implemented with the Statistical Parametric Mapping (SPM12) software and the DPABI toolbox [43]; details are provided in Appendix B.

4.2. Evaluation of FCM-VAE with Data Augmentation

We begin by quantitatively assessing the quality of synthetic FC matrices generated by FCM-VAE. Specifically, for each site in each dataset, we split the data into 70% training and 30% testing, generate a synthetic FC graph for every training sample, and augment the training set with an equal number of synthetic samples. A GCN classifier [19] is trained on the augmented data and evaluated on the held-out test split. We compare FCM-VAE against three state-of-the-art brain network generative models, including ReGate [26], BrainNetGAN [22], and GR-SPD-GAN [38], and also report results without augmentation. Performance is measured using accuracy, precision, recall, and F1-score. As shown in Table 1, FCM-VAE consistently yields the best or second-best performance across all datasets, particularly in classification accuracy, with GR-SPD-GAN ranking second overall.

Additional results using other classifiers, such as GAT [42], GraphSAGE [13], GAE [18], and GraphTransformer [10], as well as visualizations of synthesized FC matrices produced by different approaches (e.g., FCM-VAE and GR-SPD-GAN), are provided in Appendix D.

4.3. Evaluation of FORGE

In this section, we systematically evaluate the proposed continual learning framework FORGE under a unified ex-

Table 1. FC classification results with data augmentation using different graph generation methods. Columns report ACC, PRE, REC, and F1. Bold numbers denote the best results, and underlined numbers indicate the second-best.

Classifier	Gen. Method	ASD				BSNIP				MDD			
		ACC	PRE	REC	F1	ACC	PRE	REC	F1	ACC	PRE	REC	F1
GCN	No Augmentation	0.762	0.748	0.772	0.758	0.688	0.733	0.456	0.502	0.756	0.733	0.699	0.700
	ReGate	0.724	0.688	<u>0.806</u>	0.740	0.639	0.618	0.318	0.361	0.724	0.701	0.725	0.676
	BrainNetGan	0.758	0.757	0.752	0.747	0.686	<u>0.747</u>	<u>0.458</u>	0.488	0.748	<u>0.707</u>	0.745	0.712
	GR-SPD-GAN	<u>0.780</u>	<u>0.761</u>	0.812	0.780	<u>0.703</u>	0.785	0.452	<u>0.513</u>	<u>0.767</u>	0.666	0.866	0.745
	FCM-VAE (Ours)	0.783	0.776	0.775	<u>0.771</u>	0.727	0.727	0.547	0.597	0.787	0.733	<u>0.784</u>	<u>0.735</u>

perimental protocol. To comprehensively assess its adaptability and resilience to catastrophic forgetting in multi-task and cross-site settings, four representative categories of continual learning methods are adopted as baselines for comparison: regularization-based methods (EWC [20], SI [48]), parameter isolation approaches (PackNet [28]), rehearsal-based strategies (ER [33]), and knowledge distillation methods (LwF [24], DER [2], DER++ [2]). In addition, we also consider three modern frameworks specifically designed for continual learning on graph classification tasks, including TWP [25], UGCL [14], and PDGNNs [52]. According to the results in Table 1, GCN achieves the highest average accuracy among all candidate classifiers and is therefore adopted as the backbone for all continual learning frameworks. For each site, the data are split into a 7:3 ratio for training and testing, and we fix the replay buffer size to 256 for ER, DER, DER++, and FORGE (Ours).

To evaluate continual learning performance, we adopt two standard metrics that respectively reflect stability and plasticity: Average Anytime Accuracy (AAA) and Forgetting Rate (FOR). AAA measures the overall generalization ability across sequential tasks, while FOR quantifies the degree of catastrophic forgetting. Formally,

$$AAA = \frac{1}{T} \sum_{t=1}^T \frac{1}{t} \sum_{i=1}^t ACC_{t,i}, \quad (23)$$

$$FOR = \frac{1}{T} \sum_{i=1}^T \left(\max_{t \in \{1, \dots, T-1\}} ACC_{t,i} - ACC_{T,i} \right), \quad (24)$$

where T is the total number of tasks, $ACC_{t,i}$ denotes the accuracy on task i after completing task t , and $ACC_{T,i}$ is the final accuracy on task i after learning all tasks. To ensure fair comparison, we evaluate each method under all task orders and report the averaged performance across sequences.

From the results in Table 2, FORGE consistently achieves the best balance between stability and plasticity across all three benchmarks, obtaining the highest AAA scores and the lowest FOR values. Relative to the strongest

Table 2. Different continual learning frameworks performance on ASD, BSNIP, and MDD.

CL Method	ASD		BSNIP		MDD	
	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓
Base	0.607	0.276	0.551	0.230	0.593	0.277
DER	0.663	0.202	0.632	0.125	0.666	0.175
DER++	0.672	0.189	0.623	0.138	0.695	0.147
ER	0.637	0.240	0.597	0.170	0.648	0.176
EWC	0.596	0.193	0.569	0.190	0.607	0.202
LwF	0.646	0.222	0.622	0.142	0.639	0.186
PackNet	0.663	0.203	0.618	0.136	0.624	0.186
SI	0.668	0.175	0.655	0.137	0.648	0.176
TWP	0.674	0.166	0.639	0.147	0.678	0.180
UGCL	0.680	0.159	0.640	0.144	0.692	0.161
PDGNNs	0.711	0.131	0.659	0.128	0.705	0.139
FORGE (Ours)	0.730	0.116	0.685	0.101	0.738	0.105

baseline on each dataset, FORGE improves AAA by 2.7%, 3.9%, and 4.7% on ASD, BSNIP, and MDD, respectively, and reduces forgetting by 11.5%, 21.1%, and 24.5%. While PDGNNs stands as the second-best method overall, highlighting the benefits of topology design for mitigating catastrophic forgetting in graph-based continual learning. Meanwhile, knowledge distillation-based approaches such as DER and DER++ also exhibit competitive performance, demonstrating the strong potential of distillation strategies in preserving both stability and plasticity.

Furthermore, we visualize the averaged AAA and FOR across all task learning stages, as illustrated in Figure 4. It can be observed that FORGE maintains superior accuracy throughout the continual learning process. In contrast, other methods suffer from noticeable performance drops, particularly in later stages, revealing their limited ability to retain previous knowledge and adapt stably over time.

4.4. Ablation Study and Sensitivity Analysis

To further investigate the contribution of each component in our framework, we conduct an ablation study as summarized in Table 3. Removing different components from FORGE leads to distinct performance degradations. The absence of GKD results in higher forgetting and lower AAA, while removing LKD further weakens task-specific

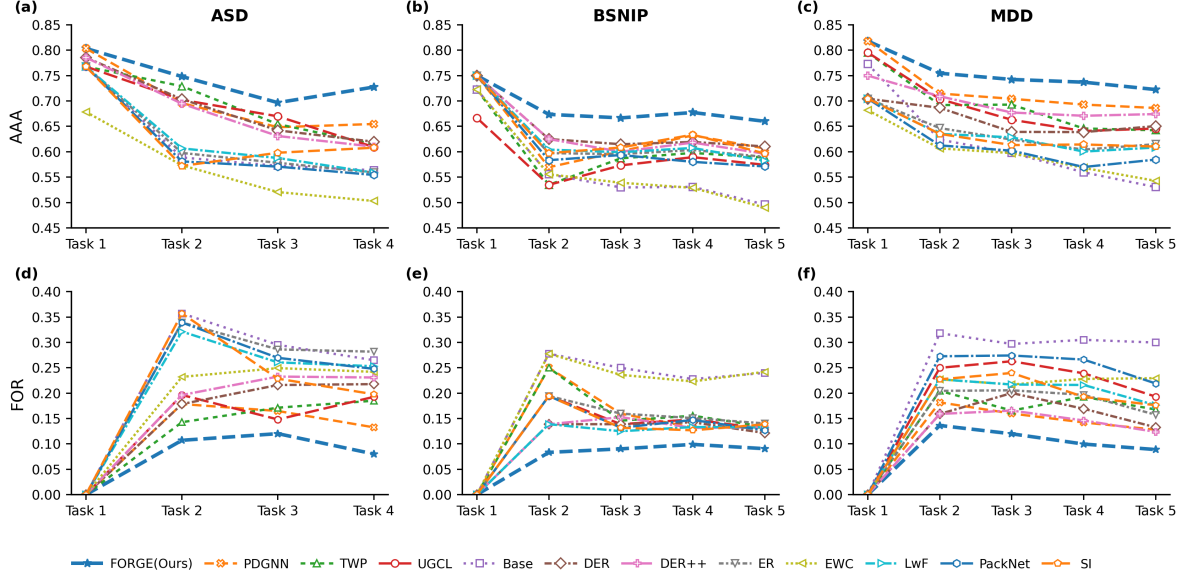


Figure 4. Visualization of the averaged AAA and FOR across sequential tasks under a fixed task order on ASD, BSNIP, and MDD.

Table 3. Ablation study assessing the contribution of Graph Knowledge Distillation (GKD), Logits Knowledge Distillation (LKD), and Hierarchical Contextual Thompson Sampling (HCTS) to the overall performance of the FORGE framework.

Method	ASD		BSNIP		MDD	
	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓
Full Framework	0.730	0.116	0.685	0.101	0.738	0.105
w/o GKD	0.683	0.142	0.637	0.131	0.691	0.141
w/o LKD	0.681	0.151	0.638	0.125	0.695	0.133
w/o HCTS	0.676	0.163	0.634	0.137	0.685	0.144

retention. Eliminating the HCTS causes the most severe drop, yielding the highest forgetting across all datasets.

Table 4. Performance of continual learning methods under different buffer sizes on the ASD dataset.

Buffer Size	FORGE (Ours)		ER		DER		DER++	
	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓
128	0.685	0.157	0.601	0.275	0.628	0.221	0.627	0.229
256	0.730	0.116	0.637	0.240	0.663	0.202	0.672	0.189
384	0.719	0.129	0.642	0.215	0.658	0.217	0.669	0.210

To assess the impact of memory capacity, we evaluate FORGE and competing methods using buffer under different buffer sizes, as reported in Table 4. FORGE consistently achieves the best performance across all settings, with the advantage most evident in the low-buffer regime. This robustness stems from HCTS, which effectively selects more informative and diverse synthetic samples to maximize the utility of the constrained buffer. Interestingly, performance

declines slightly as the buffer size increases. This may be because a larger buffer size introduces more redundant or less informative generated samples.

τ	Base		DER++		PDGNN		FORGE (Ours)	
	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓	AAA↑	FOR↓
$\tau = 0.2$	0.595	0.306	0.655	0.200	0.709	0.123	0.727	0.109
$\tau = 0.3$	0.589	0.273	0.670	0.197	0.707	0.125	0.721	0.114
$\tau = 0.4$	0.607	0.276	0.672	0.189	0.711	0.131	0.730	0.116
$\tau = 0.5$	0.613	0.293	0.680	0.213	0.716	0.143	0.735	0.120

Table 5. Sensitivity to FC threshold τ on ASD.

We further examine the sensitivity of our framework to different threshold values of τ in graph construction. As shown in Table 5, FORGE demonstrates strong robustness to different τ values, consistently yielding the best AAA and the lowest FOR. This suggests that the observed gains are stable rather than tied to a particular threshold setting. Additional ablation results are provided in Appendix E.

5. Conclusion

We introduced FORGE, the first continual learning framework for fMRI-based brain disorder diagnosis. At its core, FCM-VAE is a structure-aware generative model that synthesizes high-quality functional connectivity matrices, providing a viable solution for privacy-preserving replay in clinical environments. FORGE further stabilizes learning through dual-level distillation and adaptive sampling, ensuring knowledge retention as new sites arrive. Experiments on three large-scale neuroimaging datasets demonstrate consistent improvements in accuracy and substantial reductions in forgetting compared with state-of-the-art approaches.

References

- [1] Fatemeh Amrollahi, Supreeth P Shashikumar, Andre L Holder, and Shamim Nemati. Leveraging clinical data across healthcare institutions for continual learning of predictive risk models. *Scientific reports*, 12(1):8380, 2022. 1
- [2] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: a strong, simple baseline. *Advances in neural information processing systems*, 33:15920–15930, 2020. 2, 7, 6
- [3] Antonio Carta, Andrea Cossu, Federico Errica, and Davide Bacciu. Catastrophic forgetting in deep graph networks: an introductory benchmark for graph classification. *arXiv preprint arXiv:2103.11750*, 2021. 2
- [4] Arslan Chaudhry, Marc’Aurelio Ranzato, Marcus Rohrbach, and Mohamed Elhoseiny. Efficient lifelong learning with a-GEM. In *International Conference on Learning Representations*, 2019. 2
- [5] Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet K Dokania, Philip HS Torr, and Marc’Aurelio Ranzato. On tiny episodic memories in continual learning. *arXiv preprint arXiv:1902.10486*, 2019. 1
- [6] Brett A Clementz, David A Parker, Rebekah L Trotti, Jennifer E McDowell, Sarah K Keedy, Matcheri S Keshavan, Godfrey D Pearlson, Elliot S Gershon, Elena I Ivleva, Ling-Yu Huang, et al. Psychosis biotypes: replication and validation from the b-snip consortium. *Schizophrenia bulletin*, 48(1):56–68, 2022. 2, 6
- [7] Pradnya Desai, Phung Lai, NhatHai Phan, and My T Thai. Continual learning with differential privacy. In *International Conference on Neural Information Processing*, pages 334–343. Springer, 2021. 1
- [8] Adriana Di Martino, Chao-Gan Yan, Qingyang Li, Erin Denio, Francisco X Castellanos, Kaat Alaerts, Jeffrey S Anderson, Michal Assaf, Susan Y Bookheimer, Mirella Dapretto, et al. The autism brain imaging data exchange: towards a large-scale evaluation of the intrinsic brain architecture in autism. *Molecular psychiatry*, 19(6):659–667, 2014. 1, 2, 6
- [9] Tiehang Duan, Zhenyi Wang, Fang Li, Gianfranco Doretto, Donald A Adjeroh, Yiyi Yin, and Cui Tao. Online continual decoding of streaming eeg signal with a balanced and informative memory buffer. *Neural Networks*, 176:106338, 2024. 2
- [10] Vijay Prakash Dwivedi and Xavier Bresson. A generalization of transformer networks to graphs. *arXiv preprint arXiv:2012.09699*, 2020. 6, 4
- [11] Selene Gallo, Ahmed El-Gazzar, Paul Zhutovsky, Rajat M Thomas, Nooshin Javaheripour, Meng Li, Lucie Bartova, Deepti Bathula, Udo Dannlowski, Christopher Davey, et al. Functional connectivity signatures of major depressive disorder: machine learning analysis of two multicenter neuroimaging studies. *Molecular Psychiatry*, 28(7):3013–3022, 2023. 1
- [12] Aris Gkoulalas-Divanis and Grigorios Loukides. Introduction to medical data privacy. In *Medical data privacy handbook*, pages 1–14. Springer, 2015. 1
- [13] Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. *Advances in neural information processing systems*, 30, 2017. 6, 4
- [14] Thanh Duc Hoang, Do Viet Tung, Duy-Hung Nguyen, Bao-Sinh Nguyen, Huy Hoang Nguyen, and Hung Le. Universal graph continual learning. *arXiv preprint arXiv:2308.13982*, 2023. 2, 7
- [15] Peter D Hoff, Adrian E Raftery, and Mark S Handcock. Latent space approaches to social network analysis. *Journal of the American Statistical Association*, 97(460):1090–1098, 2002. 5
- [16] Jeremy Kawahara, Colin J Brown, Steven P Miller, Brian G Booth, Vann Chau, Ruth E Grunau, Jill G Zwicker, and Ghassan Hamarneh. Brainnetcn: Convolutional neural networks for brain networks; towards predicting neurodevelopment. *NeuroImage*, 146:1038–1049, 2017. 1
- [17] Junsu Kim, Hoseong Cho, Jihyeon Kim, Yihalem Yimolal Tiruneh, and Seungryul Baek. Sddgr: Stable diffusion-based deep generative replay for class incremental object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28772–28781, 2024. 2
- [18] Thomas N Kipf and Max Welling. Variational graph auto-encoders. *arXiv preprint arXiv:1611.07308*, 2016. 6, 4
- [19] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016. 6, 4
- [20] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017. 1, 2, 7
- [21] Jihoon Ko, Shinhwan Kang, Taehyung Kwon, Heechan Moon, and Kijung Shin. Begin: Extensive benchmark scenarios and an easy-to-use framework for graph continual learning. *ACM Transactions on Intelligent Systems and Technology*, 16(1):1–22, 2025. 2
- [22] Chao Li, Yiran Wei, Xi Chen, and Carola-Bibiane Schönlieb. Brainnetgan: Data augmentation of brain connectivity using generative adversarial network for dementia classification. In *MICCAI Workshop on Deep Generative Models*, pages 103–111. Springer, 2021. 6, 3, 4
- [23] Xiaoxiao Li, Yufeng Gu, Nicha Dvornek, Lawrence H Staib, Pamela Ventola, and James S Duncan. Multi-site fmri analysis using privacy-preserving federated learning and domain adaptation: Abide results. *Medical image analysis*, 65: 101765, 2020. 1
- [24] Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2935–2947, 2017. 2, 7
- [25] Huihui Liu, Yiding Yang, and Xinchao Wang. Overcoming catastrophic forgetting in graph neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, pages 8653–8661, 2021. 2, 7

- [26] Meimei Liu, Zhengwu Zhang, and David B Dunson. Graph auto-encoding brain networks with applications to analyzing large-scale brain imaging datasets. *Neuroimage*, 245: 118750, 2021. [5](#), [6](#), [3](#), [4](#)
- [27] Nikos K Logothetis. What we can do and what we cannot do with fmri. *Nature*, 453(7197):869–878, 2008. [2](#)
- [28] Arun Mallya and Svetlana Lazebnik. Packnet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pages 7765–7773, 2018. [1](#), [2](#), [7](#)
- [29] Michael McCloskey and Neal J Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation*, pages 109–165. Elsevier, 1989. [1](#)
- [30] German I Parisi, Ronald Kemker, Jose L Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural networks*, 113:54–71, 2019. [1](#)
- [31] Mark Plitt, Kelly Anne Barnes, and Alex Martin. Functional connectivity classification of autism identifies highly predictive brain features but falls short of biomarker standards. *NeuroImage: Clinical*, 7:359–366, 2015. [1](#)
- [32] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H Lampert. icarl: Incremental classifier and representation learning. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pages 2001–2010, 2017. [2](#)
- [33] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy Lillicrap, and Gregory Wayne. Experience replay for continual learning. *Advances in neural information processing systems*, 32, 2019. [7](#)
- [34] Andrei A Rusu, Neil C Rabinowitz, Guillaume Desjardins, Hubert Soyer, James Kirkpatrick, Koray Kavukcuoglu, Razvan Pascanu, and Raia Hadsell. Progressive neural networks. *arXiv preprint arXiv:1606.04671*, 2016. [2](#)
- [35] Manish Saggar, Olaf Sporns, Javier Gonzalez-Castillo, Peter A Bandettini, Gunnar Carlsson, Gary Glover, and Allan L Reiss. Towards a new approach to reveal dynamical organization of the brain using topological data analysis. *Nature communications*, 9(1):1399, 2018. [2](#)
- [36] Yufei Shi, Shishi Tang, Yuxuan Li, Zhipeng He, Shengsheng Tang, Ruixuan Wang, Weishi Zheng, Ziyi Chen, and Yi Zhou. Continual learning for seizure prediction via memory projection strategy. *Computers in Biology and Medicine*, 181:109028, 2024. [2](#)
- [37] Yiping Song, Juhua Zhang, Zhiliang Tian, Yuxin Yang, Minlie Huang, and Dongsheng Li. Llm-based privacy data augmentation guided by knowledge distillation with a distribution tutor for medical text classification. *arXiv preprint arXiv:2402.16515*, 2024. [2](#)
- [38] Yee-Fan Tan, Fuad Noman, Raphaël C-W Phan, Hernando Ombao, and Chee-Ming Ting. Graph-regularized manifold-aware conditional wasserstein gan for brain functional connectivity generation. *Human Brain Mapping*, 46(12):e70322, 2025. [6](#), [3](#), [4](#)
- [39] Zonggui Tian, Du Zhang, and Hong-Ning Dai. Continual learning on graphs: A survey. *arXiv preprint arXiv:2402.06330*, 2024. [2](#)
- [40] Nathalie Tzourio-Mazoyer, Brigitte Landeau, Dimitri Papathanassiou, Fabrice Crivello, Octave Etard, Nicolas Delcroix, Bernard Mazoyer, and Marc Joliot. Automated anatomical labeling of activations in spm using a macroscopic anatomical parcellation of the mni mri single-subject brain. *Neuroimage*, 15(1):273–289, 2002. [2](#)
- [41] Guido M Van de Ven and Andreas S Tolias. Three scenarios for continual learning. *arXiv preprint arXiv:1904.07734*, 2019. [1](#)
- [42] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, and Yoshua Bengio. Graph attention networks. *arXiv preprint arXiv:1710.10903*, 2017. [6](#), [4](#)
- [43] Chao-Gan Yan, Xin-Di Wang, Xi-Nian Zuo, and Yu-Feng Zang. Dpabi: data processing & analysis for (resting-state) brain imaging. *Neuroinformatics*, 14(3):339–351, 2016. [6](#)
- [44] Chao-Gan Yan, Xiao Chen, Le Li, Francisco Xavier Castellanos, Tong-Jian Bai, Qi-Jing Bo, Jun Cao, Guan-Mao Chen, Ning-Xuan Chen, Wei Chen, et al. Reduced default mode network functional connectivity in patients with recurrent major depressive disorder. *Proceedings of the National Academy of Sciences*, 116(18):9078–9083, 2019. [2](#), [6](#)
- [45] Chengxuan Ying, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. Do transformers really perform badly for graph representation? *Advances in neural information processing systems*, 34: 28877–28888, 2021. [4](#)
- [46] Jaehong Yoon, Eunho Yang, Jeongtae Lee, and Sung Ju Hwang. Lifelong learning with dynamically expandable networks. In *International Conference on Learning Representations*, 2018. [2](#)
- [47] Shikang Yu, Jiachen Chen, Hu Han, and Shuqiang Jiang. Data-free knowledge distillation via feature exchange and activation region constraint. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24266–24275, 2023. [2](#)
- [48] Friedemann Zenke, Ben Poole, and Surya Ganguli. Continual learning through synaptic intelligence. In *International conference on machine learning*, pages 3987–3995. Pmlr, 2017. [7](#)
- [49] Ding Zhang, Jane Downer, Can Chen, and Ren Wang. Efficient and robust continual graph learning for graph classification in biology. *IEEE Transactions on Signal and Information Processing over Networks*, 2025. [2](#)
- [50] Qifan Zhang, Yunhui Guo, and Yu Xiang. Continual distillation learning: Knowledge distillation in prompt-based continual learning. *arXiv preprint arXiv:2407.13911*, 2024. [2](#)
- [51] Xikun Zhang, Dongjin Song, and Dacheng Tao. Hierarchical prototype networks for continual graph representation learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4):4622–4636, 2022. [2](#)
- [52] Xikun Zhang, Dongjin Song, Yixin Chen, and Dacheng Tao. Topology-aware embedding memory for continual learning on expanding networks. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 4326–4337, 2024. [2](#), [7](#)
- [53] Zhiyu Zhang, Wei Chen, Youfang Lin, and Huaiyu Wan. A generative adaptive replay continual learning model for

- temporal knowledge graph reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10964–10977, 2025. [2](#)
- [54] Kaizhong Zheng, Shujian Yu, Baojuan Li, Robert Jenssen, and Badong Chen. Brainib: Interpretable brain network-based psychiatric diagnosis with graph information bottleneck. *IEEE Transactions on Neural Networks and Learning Systems*, 36(7):13066–13079, 2024. [1](#), [6](#)
- [55] Fan Zhou and Chengtai Cao. Overcoming catastrophic forgetting in graph neural networks with experience replay. In *Proceedings of the AAAI conference on artificial intelligence*, pages 4714–4722, 2021. [2](#)
- [56] Yangxuan Zhou, Sha Zhao, Jiquan Wang, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. Brainuicl: An unsupervised individual continual learning framework for eeg applications. In *The Thirteenth International Conference on Learning Representations*, 2025. [2](#)